

Data Dictionary

Zero-Inflated Negative Binomial (ZINB) Latent-Class Trajectories

File: zinb.sas

Last updated: July 21, 2026

Package: Traj2 (prototype, under QA)

1. Overview

This file fits zero-inflated negative binomial latent-class (growth-mixture) trajectory models in PROC NLMIXED. It extends the ZIP model with a class-specific NB2 dispersion parameter ($k = \exp(\log k)$) to accommodate overdispersion ($\text{Var} = \mu + \mu^2/k$), common in high-utilization Medicare count data. Starting values are set via %let lines; unset parameters default to 0.

2. Datasets

Name	Description
sim_long / sim_wide	Long / wide outputs of the ZINB simulator (%sim_data), which also builds BASE_FILE_SRS internally.
BASE_FILE_SRS	Wide input: BENE_ID, SUM_Q1 . . SUM_Q12, quar1 . . quar12 (values 1..T).
pe_zinb	ParameterEstimates from the fit (default name).
fit_zinb	FitStatistics from the fit (default name).
traj_zinb / mix_zinb	Per-class ZINB mean curves and the mixture-mean series.

3. Macro Arguments (%ct_zinb_nlmixed)

Name	Type / Default	Description
data	name, BASE_FILE_SRS	Input wide-format dataset.
id	name, BENE_ID	Subject identifier.
yvars	varlist, SUM_Q1-SUM_Q12	Repeated-measure count outcomes.
nclass	integer, 5	Number of latent classes K.
class_labels	list, A B C D E	Space-separated labels; must match nclass (checked; mismatch aborts the macro).
order	integer, 2	Mean-curve polynomial degree.
p_order	integer, 0	Zero-inflation logit polynomial degree.
T	integer, 12	Number of time points.
tech	name, newrap	Optimization technique.
maxiter	integer, 500	Maximum iterations.
pe_out	name, pe_zinb	Output dataset for parameter estimates.
fit_out	name, fit_zinb	Output dataset for fit statistics.

Name	Type / Default	Description
bounds	text, optional	Parameter bounds for stability.

4. Key Variables and Parameters

Name	Type	Description
BENE_ID	Num	Subject identifier.
Y[T]	Array	Count outcome over time inside the model.
alpha0_[label]	Num	Class-membership logit (class A is reference).
beta0/1/2/3_[label]	Num	Log-link mean polynomial coefficients.
gamma0/1/2/3_[label]	Num	Zero-inflation logit polynomial coefficients.
logk_[label]	Num	Log NB dispersion; size $k = \exp(\log k) > 0$.
k_[label]	Num	NB2 size parameter for the class.
mu_[label][T]	Array	Class mean count trajectory, $\exp(\eta)$.
logitp_[label][T]	Array	Class zero-inflation logit over time.
p_[label][T]	Array	Class structural-zero probability over time.
logNB / logNB0	Num	NB2 log pmf and NB mass at zero.
pi_[label][T]	Num	Per-time log-likelihood contribution.
w_[label]	Num	Class mixing weight (softmax).
prod_[label]	Num	Summed per-class log-likelihood across time.
ll	Num	Mixture log-likelihood (log-sum-exp) for the model.

5. Starting-Value Macro Variables (Section [C])

Name	Description
alpha0_B, alpha0_C, ...	Mixing logits; class A has none (reference).
beta0_A, beta1_A, ...	Mean-curve coefficients per class.
gamma0_A, gamma0_B, ...	Zero-inflation logits per class.
logk_A, logk_B, ...	NB dispersion ($k = \exp(\log k)$); larger k = closer to Poisson.
(any unset parameter)	Defaults to 0 in the PARMS statement.

6. Macros

Name	Purpose
%sim_data	ZINB simulator (Gamma-Poisson mixture); also builds BASE_FILE_SRS.

Name	Purpose
<code>%_ct_nwords / %_ct_abort</code>	Word count and safe stop for the nclass check.
<code>%_emit_parm</code>	Emits one PARMS entry from a <code>%let</code> value or 0 default.
<code>%_ct_array_from_list</code>	Declares the outcome array.
<code>%_ct_declare_mu_arrays / _pi_ / _zip_</code>	Declare mu, pi, logitp/p arrays per class.
<code>%_ct_parms_all</code>	Builds the single PARMS statement (alphas, betas, gammas, logk).
<code>%_ct_fill_mu_poly</code>	Fills class mean trajectories.
<code>%_ct_fill_zip_poly</code>	Fills class zero-inflation trajectories.
<code>%_ct_accumulate_zinb_ll</code>	Accumulates per-class ZINB log-likelihood.
<code>%_ct_mixture_ll</code>	Mixes classes via softmax weights (log-sum-exp).
<code>%ct_zinb_nlmixed</code>	Main fit macro.
<code>%ct_zinb_plots</code>	Class curves and mixture-mean overlay.

7. Notes and Assumptions

- Prototype: interfaces and behavior may change before release.
- NB2 mean μ , variance $\mu + \mu^2/k$; as k grows, ZINB approaches ZIP.
- `nclass` must equal the number of `class_labels`, or the macro aborts.
- Class 1 (label A) is the mixing reference; no α_0 for it.
- Missing Y values are skipped in the likelihood.
- Warm starts for $\log k$ near 0 ($k = 1$) are reasonable; adjust if convergence is slow.